

What is Git?

Git is the most widely used distributed version control system.

Key characteristics:

- Tracks changes in source code and other files
- Works locally on your computer
- Supports branching and merging
- Enables collaboration through remote repositories

Created by Linus Torvalds in 2005.

What is Version Control?

Version control is a system that records changes to files over time so you can:

- See what changed and when
- Revert to previous versions
- Compare different versions
- Collaborate with multiple developers
- Prevent accidental loss of work

Think of it as a detailed "undo history" for your entire project.

Core Git Concepts

Concept	Description
Repository (Repo)	A project tracked by Git
Commit	A snapshot of changes at a specific point in time
Branch	An independent line of development
Merge	Combining changes from different branches
Clone	Copying a repository to your computer
Push	Uploading local commits to a remote repository
Pull	Downloading changes from a remote repository
Staging Area	Temporary area where changes are prepared for commit

What is Git?

Free & Open Source **Version Control System**



tools that help to tracks changes in code

- 1) track history
- 2) help to collaborate

What is Github?

Website where we host **repositories** online

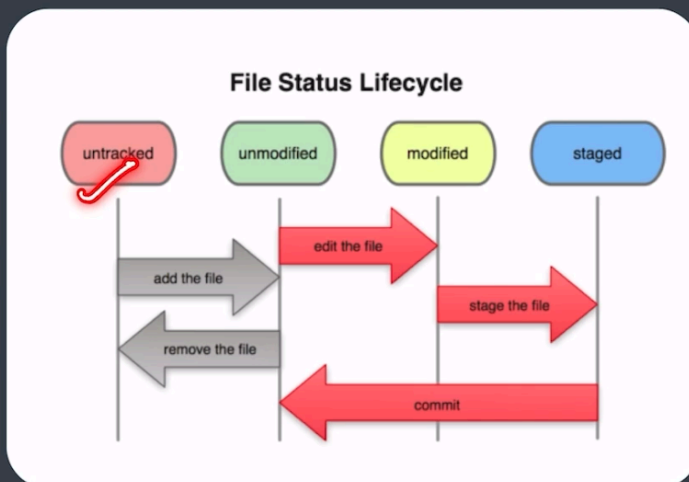


Basic Commands

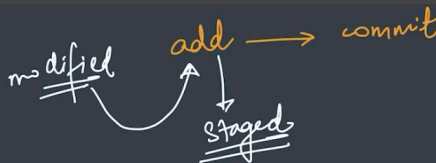
- **Clone** - Cloning a repository on our local machine
- **status** - displays the state of the code

`git clone <- some link ->`

`git status`



Basic Commands



- **add** - adds new or changed files in your working directory to the Git staging area.
- **commit** - it is the record of change

`git add <- file name ->`

`git commit -m "some message"`

- **push** - upload local repo content to remote repo

`git push origin main`

Step 1: Check the status

```
</> Bash  
git status
```



Step 2: Commit (if you haven't already)

```
</> Bash  
git add .  
git commit -m "Updated README"
```



Step 3: Push to GitHub

```
</> Bash  
git push origin main
```



If your default branch is `master` instead of `main`, use:

```
</> Bash  
git push origin master
```



Basic Commands

- **init** - used to create a new git repo *git init*

git remote add origin <- link ->

git remote -v (to verify remote)

git branch (to check branch)

git branch -M main (to rename branch)

git push origin main